KATHIR V



PERSONAL DETAILS

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PROFILE

Bioinformatics graduate with hands-on experience in Python and Machine Learning, focused on developing data-driven solutions for biological and healthcare applications. Experienced in feature engineering, model development, evaluation, and data analysis using Python, scikit-learn and TensorFlow. Built ML projects involving protein sequence classification and healthcare audio/image data, with a strong interest in applying AI/ML to real-world healthcare problems.

EDUCATION

Bachelor of Technology in Bioinformatics — Aug 2022 - May 2026
SASTRA DEEMED UNIVERSITY, THANJAVUR

- CGPA: 7.46/10

SKILLS

- Python
- SQL
- Machine Learning
- scikit-learn
- TensorFlow
- Keras
- PyTorch
- NumPy
- Pandas
- Data Analysis
- Feature Engineering
- Bioinformatics
- Protein Sequence Analysis
- k-mer Representation
- Linux (data processing, scripting for biodata analysis)

PROJECTS

Lung Sound Classification System & Android App (Python, ML, Android Studio)

- Developed an ML-based system to classify respiratory diseases from lung sound recordings.
- Built and evaluated Random Forest and MLP models, achieving over 91% accuracy.
- Integrated the trained ML model into an Android application for prediction from recorded lung sounds.
- Implemented the workflow from audio input and preprocessing to model prediction and mobile deployment.

App link: https://play.google.com/store/apps/details?id=com.gs.safebreath

Protein Family Classification using Physicochemical Properties (Python, Scikit-learn)

- Developed an ML pipeline for classifying protein sequences into families using physicochemical properties of amino acids.
- Engineered biochemical, hydropathy and structural features and transformed sequences into property-based k-mer representations.
- Evaluated feature representations and ML models to study their effect on classification performance.
- Achieved improved performance compared with conventional k-mer frequency representation while using fewer features and reducing training time.

Blood Cell Classification using Vision Transformer (Python, PyTorch, ViT, Score-CAM)

- Developed a Vision Transformer-based model to classify RBC, WBC and platelet cell images.
- Trained the model without pretrained weights and achieved 99.18% test accuracy.
- Applied data augmentation and noise-based training.
- Used Score-CAM to visualize regions contributing to model predictions and improve interpretability.

Dog Breed Image Classification (Python, TensorFlow, MobileNetV2)

- Developed a MobileNetV2-based image classification model.
- Trained and evaluated it on 10,000+ images and achieved 92.9% test accuracy.

CERTIFICATES

- AI & Machine Learning and Data Science - Udemy, 2024
- Generative AI & Prompt Engineering - HCLTech, 2025